COURSE OUTLINE

Course Code: CSE 106

Course Title: Data Structures and Algorithms I Sessional

Level/Term: 1/II **Section: A/B/C**

Academic Session: 2023-24

Course Teacher(s):

Name	Email and Telephone (Optional)
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Course Outline:

Laboratory works covering the following topics: Introduction to algorithms; Asymptotic analysis: growth of functions, O, Ω, Θ, o, ω notations; Correctness proof of algorithms; Analysis of algorithms: Master theorem, etc.; Elementary data structures: arrays, linked lists, stacks, queues, trees, and tree traversals, graphs and graph representations, heaps, binary search trees; Graph traversals: DFS, BFS, applications of DFS and BFS; Methods for the design of

efficient algorithms: divide and conquer, greedy methods, dynamic programming; Sorting: sorting algorithms, sorting in linear time; Lower bound theory; Data structures for set operations.

Assessment

Attendance and Practice Performance:	5-10%
Lab and Home Assignments:	60-70%
Quiz:	25-30%

**** All weights are subject to change.**

Text and Reference books:

Same as CSE 105.

Rules (***STRICT***)

1. Do not copy code in online or offline. You will get penalized when you are caught (and you will).
2. When submitting an offline / home assignment, strictly follow the steps specified in the assignment description to prepare and submit your file(s).
3. For offline assignments, you have to submit your file(s) via email (details to be mentioned in the assignment description). Additionally, on the evaluation day, you must be able to explain your code properly. You must bring your code on that day too.
4. You can use online resources/tools to guide your solution. However, copying solutions from any such resources is strictly prohibited. You are expected to implement your solutions, clearly describe them during the evaluations, and propose relevant modifications if asked.

Weekly schedule:

Week	Topic
Week 1	Discussion on subsequent course plan
Week 2	**Online:** Simple Array-based problems Assignment 1 Declaration: Array List and Linked List
Week 3	**Evaluation** of Assignment 1 with **Online** Assignment 2 Declaration: Stack
Week 4	**Evaluation** of Assignment 2 with **Online** Assignment 3 Declaration: Queue
Week 5	**Evaluation** of Assignment 3 with **Online** Assignment 4 Declaration: Binary Search Tree
Week 6	Discussion or Practice or Online
Break	
Week 7	**Evaluation** of Assignment 4 with **Online**
Week 8	Assignment 5 Declaration: Heaps
Week 9	**Evaluation** of Assignment 5 with **Online** Assignment 6 Declaration: Graphs
Week 10	**Evaluation** of Assignment 6 with **Online** Assignment 7 Declaration: Divide and Conquer and Greedy Algorithms
Week 11	**Evaluation** of Assignment 7 with **Online** Assignment 8 Declaration: Dynamic Programming
Week 12	Evaluation of Assignment 8 with Online
Week 13	**Online** with problem-solving
Week 14	**Quiz 1**